

Dear Editor,

Please find our enclosed manuscript “**Genomic analyses reveal distinct demographic history and identify genetic changes underlying adaption, speciation and domestication in ducks**”, which we wish to publish in *PNAS*.

Oct. 5, 2018

Recently, more and more research has found that traditional domestication hypothesis is incompatible with genetic data, although phylogenetic affinities of domestic animals have been found, the direct wild progenitor of domestic animals are hardly to be identified. In this article, through reveal the evolutionary history of ducks, we demonstrate that both the traditional hypotheses that domestic duck are origin from mallard duck or the descendant of mallard and spot-billed ducks are incompatible with genetic data. Instead, there is a very deep time depth for the MRCA of wild and domestic ducks, which is longer than the divergence time between mallard and spot-billed. We demonstrate the formation of the genetic basis for the remarkable phenotypic variations between domestic and wild ducks, such as stressful nature selection may grant wild duck long term flight and migration ability in ancient, recent selection in domestic ducks affected genes involving behavior and growth. Thus making duck which is one of the principal natural hosts of influenza A viruses and the most economical important waterfowl, to be an excellent model to shed light the understanding on domestication process.

Here, we firstly present the whole-genome sequences of two wild duck species and provide genomic sequencing data of eight domestic duck breeds and eight wild duck species, together with two outgroups (Muscovy duck and Taihu goose). We further perform demography and selection analyses in wild and domestic ducks, and provide functional support for candidate mutations in key genes. Our finding may help to understand how animals are domesticated and why, in some case (dog), none of the extent wild populations are more related to domestic forms. What is the different mechanism between speciation process and domestication process, which both involving isolation and producing new species (or breeds), and may help us to understand the origins of biodiversity. Whole genomic sequence of mallard and spot-billed ducks who have only tens of thousands year divergence time that is largely shorter than previous thought, could serve as an important material for revealing what happened in the very beginning stage of speciation process, especially which process drive the evolution of genomic divergence.

We feel our work here will be of interest to the broad readership of *PNAS* as it reveals new insights in the pattern of evolution and selection of domestic ducks and wild ducks, and provides new data resources and analyses that have an impact on genetic architecture for speciation divergence and artificial selection of domestic animals. Our submission includes one main text, five figures, one supplemental information file

and eight supplemental tables.

We are very much looking forward to hearing a positive response from you.

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